Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2025
Team ID	LTVIP2026TMIDS54144
Project Name	ShopSmart – Digital Grocery Web Application
Maximum Marks	4 Marks

Presentation Layer:
- Angular frontend for a dynamic Single Page Application (SPA).

Application Layer:
- Node.js + Express server to handle business logic and RESTful API routes.

Data Layer:
- MongoDB for flexible, scalable NoSQL document storage.

External Interface:
- Stripe Payment API for secure online transactions.

Infrastructure:
- Local Development environment for testing.
- Cloud Deployment utilizing MongoDB Atlas for database hosting and Render/Netlify for application hosting.

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/

Guidelines:

Include all the processes (As an application logic / Technology Block)
Provide infrastructural demarcation (Local / Cloud)

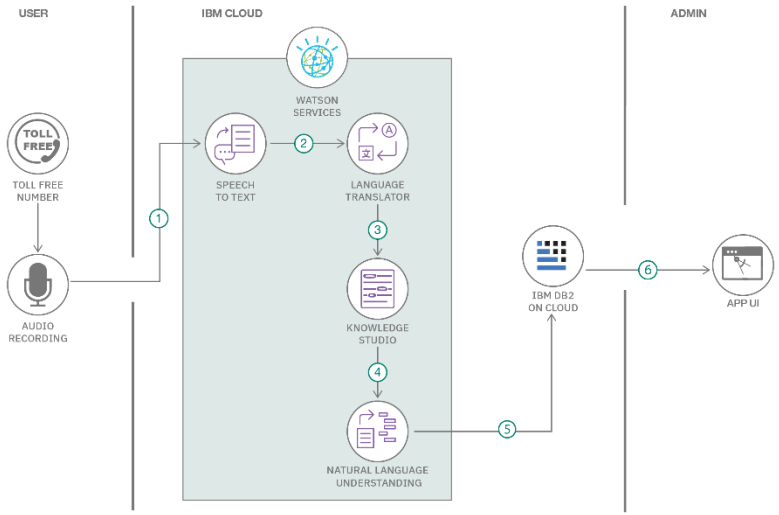

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web UI for customers to browse groceries	Angular, Bootstrap
2	Application Logic	Handles API requests and business logic	Node.js, Express
3	Authentication Logic	Secures APIs and limits access	JWT, bcrypt
4	Database	Stores products, users, and orders	MongoDB
5	Cloud Database	Hosted database cluster	MongoDB Atlas
6	File Storage	Storing configuration and uploads	Local storage / Cloud storage
7	External API	Payment gateway integration	Stripe API
8	Infrastructure	Deployment and hosting environments	Local + Cloud hosting

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Use of community-backed robust frameworks	Angular, Node.js, Express, MongoDB
2	Security	Implementation of secure auth mechanisms to protect routes	JWT, bcrypt hashing, CORS, Environment variables
3	Scalable Architecture	Clear separation of concerns enabling vertical/horizontal scaling	3-tier architecture, REST APIs
4	Availability	Reliable hosting providers ensuring the app is always accessible	Cloud hosting with 99% uptime
5	Performance	Ensuring fast load times and quick query resolution	Optimized queries, MongoDB indexing, caching possibility

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d